No-Manipulation Conditions for Heterogeneous Power-Law Cross-Impact Kernels: Sharp Characterisation and Boundary Analysis

Abstract

We derive necessary and sufficient conditions for the absence of price manipulation in a multi-asset market with transient cross-impact governed by a power-law propagator. For the singular kernel $H_{ij}(t) = c_{ij}\, t^{-\beta_{ij}}$ with heterogeneous decay exponents $\beta_{ij} \in (0, \frac{1}{2})$, we show that price manipulation is precluded (simultaneously for all time horizons $T > 0$) if and only if (a) the cross-impact exponents satisfy the *mean-exponent condition* $\beta_{ij} = (\beta_{ii} + \beta_{jj})/2$, and (b) the Gamma-weighted matrix $\widetilde{C} := [c_{ij}\, \gamma(\beta_{ij})]$ is nonnegative definite, where $\gamma(\beta) = 2\,\Gamma(1-\beta)\sin(\pi\beta/2)$.

For the regularised kernel $H_{ij}^{(t_0)}(t) = c_{ij}\,(t + t_0)^{-\beta_{ij}}$, we discover an asymmetric necessity structure: when cross-impact decays *faster* than the mean of the self-impact exponents, the mean-exponent condition remains necessary; when it decays *slower*, an explicit t_0-dependent admissible region exists, which shrinks to zero as $t_0 \to 0^+$, recovering the singular kernel's necessity. The proof rests on a spectral asymptotic analysis of incomplete Gamma functions.

We provide a comprehensive boundary analysis: the permanent-impact limit $\beta \to 0^+$ recovers the Kyle (1985) condition $C \geq 0$; singular $\widetilde{C}$ with heterogeneous exponents provides a structural safety margin against zero-cost round trips; and the large-N scaling reveals that positive cross-impact is scalable ($c_{\max} = 1$ independent of N) while negative cross-impact vanishes as $O(1/(N-1))$.

Keywords: Cross-impact, price manipulation, power-law propagator, nonnegative definiteness, transient impact, Volterra kernel, regularised kernel.
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1 Introduction

The execution of large orders moves prices. In multi-asset markets, trading in one asset affects the prices of others—a phenomenon termed *cross-impact*. The propagator model of Bouchaud et al. [6, 7] captures this through a kernel $G(t, s) \in \mathbb{R}^{N\times N}$ that specifies how

past trades propagate into current prices: the price distortion at time t caused by a trading strategy $u \in L^2([0,T],\mathbb{R}^N)$ is

$$D_t^u = \int_0^t G(t,s)\, u_s\, ds. \tag{1}$$

For the model to be economically consistent, it must preclude *price manipulation*—that is, round-trip strategies that generate expected profit from price distortions alone. Following Huberman and Stanzl [14] and Gatheral, Schied, and Slynko [9], the no-manipulation condition reduces to the *nonnegative definiteness* of the integral operator associated with G.

For a single asset, the no-manipulation theory is well developed. Gatheral et al. [9] showed that a nonnegative, nonincreasing, convex kernel suffices; Alfonsi and Schied [3] and Schied and Zhang [23] provided further analysis for power-law propagators; Abi Jaber and Neuman [2] extended the theory to general propagator kernels via spectral methods. Jusselin and Rosenbaum [15] connected no-arbitrage conditions to the power-law exponent and rough volatility. The recent empirical study of Maitrier, Loeper, Kanazawa, and Bouchaud [17] identifies the self-impact decay exponent $\beta \approx 0.48$, confirming the power-law nature of the propagator.

For multiple assets, Abi Jaber, Neuman, and Tuschmann [1] established sufficient conditions (Theorem 2.15) under which a matrix-valued convolution kernel is nonnegative definite. Their Corollary 2.18 covers the *factorised* case $H(t) = C\,\varphi(t)$ where a single scalar decay function φ modulates a constant cross-impact matrix C. However, empirical evidence consistently shows that different asset pairs exhibit different decay exponents β_{ij} [7, 21, 12], so the factorised model is restrictive.

The no-manipulation framework for cross-impact has been approached from several complementary angles. Schneider and Lillo [24] extend Gatheral's no-dynamic-arbitrage framework to multiple assets, deriving symmetry and linearity constraints on bounded cross-impact kernels. Their matrix positive-definiteness condition is analogous to ours, but applies to bounded (non-singular) kernels without identifying the structural role of the decay exponent. Muhle-Karbe, Wang, and Webster [18] show that nonlinear price impact converges to a linear propagator with stochastic liquidity in the high-frequency limit, connecting no-manipulation conditions to market volume constraints. Kyle [16] provides the classical permanent-impact benchmark, where no-manipulation requires $C \geq 0$ without any exponent structure. Our results complement all three: we provide the sharp characterisation specifically for the power-law class, revealing that the decay exponents—not just the coefficient matrix—carry essential structural information for no-manipulation.

Contribution. We make three contributions.

First, we provide necessary and sufficient conditions for no-manipulation in the class of heterogeneous power-law kernels

$$H_{ij}(t) = c_{ij}\, t^{-\beta_{ij}}, \qquad \beta_{ij} \in (0, \tfrac{1}{2}),$$

under a natural symmetry assumption ($c_{ij} = c_{ji}$, $\beta_{ij} = \beta_{ji}$). Two conditions emerge: (a) the *mean-exponent condition* $\beta_{ij} = (\beta_{ii} + \beta_{jj})/2$, which pins cross-impact decay to the arithmetic mean of self-impact exponents; and (b) the *Gamma-weighted positive definiteness* of $\widetilde{C} = [c_{ij}\,\gamma(\beta_{ij})]$. The proof reduces the nonnegative-definiteness problem to a frequency-domain

condition via the matrix-valued Bochner–Schwartz theorem and extracts both conditions through a diagonal factorisation of the spectral matrix.

Second, for the regularised kernel $H_{ij}^{(t_0)}(t) = c_{ij}\,(t+t_0)^{-\beta_{ij}}$, we discover an *asymmetric necessity* structure (Proposition 5.5). The mean-exponent condition is necessary when cross-impact decays faster than the self-impact mean ($\delta := 2\beta_{12} - \beta_{11} - \beta_{22} < 0$), but *not* necessary when it decays slower ($\delta > 0$), with an explicit t_0-dependent admissible region that shrinks to zero as $t_0 \to 0^+$. This is the first result to distinguish the regularised kernel from the singular kernel at the level of necessity conditions, and it reveals new spectral structure in the incomplete Gamma function regime.

Third, we provide a comprehensive boundary analysis:

- The *permanent-impact limit* $\beta \to 0^+$ recovers the classical Kyle [16] condition $C \geq 0$, showing that our result smoothly interpolates between permanent and transient impact.
- When $\widetilde{C}$ is *singular*, heterogeneous exponents provide a structural safety margin against zero-cost round trips, in contrast to the homogeneous case where entire portfolio directions become costless.
- The *large-N scaling* shows an asymmetry: positive cross-impact is bounded by $c \leq 1$ independent of N, while negative cross-impact must satisfy $|c| \leq 1/(N-1) \to 0$.

Related literature. Alfonsi, Klöck, and Schied [4] pioneered the matrix-valued positive-definiteness approach in discrete time. Rosenbaum and Tomas [21] provided an abstract Bochner-type characterisation for cross-impact kernels in a Hawkes-process framework, but did not compute explicit conditions for power-law kernels. Benzaquen et al. [5] introduced the factorised cross-impact model $G(t) = C\,\varphi(t)$; Tomas, Mastromatteo, and Benzaquen [25] discussed first-principles requirements. Hey, Neuman, and Tuschmann [12] estimate cross-impact propagators nonparametrically and observe that different asset pairs have different decay rates—precisely the setting our result addresses. Huberman and Stanzl [14] established the classical permanent-impact no-manipulation condition $C \geq 0$, which our Section 6.1 recovers as a limiting case.

Organisation. Section 2 sets up the model and defines nonnegative definiteness. Section 3 states and proves the main theorem for the singular kernel. Section 4 discusses economic implications, the mean-exponent condition, and connections to empirical findings. Section 5 extends the analysis to the regularised kernel, including the asymmetric necessity result. Section 6 provides the boundary analysis (permanent-impact limit, singular $\widetilde{C}$, large-N scaling). Section 7 discusses assumption necessity and what breaks without each condition. Section 8 provides numerical illustrations and parameter sensitivity. Section 9 concludes and discusses open directions.

2 Model and Preliminaries

We consider N risky assets on a finite horizon $[0,T]$. The framework follows Abi Jaber, Neuman, and Tuschmann [1].

Definition 2.1 (Trading strategy and price distortion)**.** A *trading strategy* is a progressively measurable process $u \in \mathcal{U} := L^2([0,T],\mathbb{R}^N)$. The *price distortion* is given by (1) with

Volterra kernel $G(t,s) = \mathbf{1}_{t\geq s}\, H(t-s)$, where $H : (0,\infty) \to \mathbb{R}^{N\times N}$ is a matrix-valued convolution kernel satisfying $H \in L^2_{\mathrm{loc}}((0,\infty), \mathbb{R}^{N\times N})$.

Definition 2.2 (Transient-impact cost)**.** The *transient-impact cost* of a trading strategy u is

$$\mathcal{C}(u) := \int_0^T u_t^\top D_t^u \, dt = \int_0^T \int_0^t u_t^\top H(t-s)\, u_s \, ds\, dt = \int_0^T \int_0^T u_t^\top G(t,s)\, u_s \, ds\, dt. \tag{2}$$

This is the aggregate cost of trading against one's own price impact.

Definition 2.3 (Nonnegative definiteness)**.** A Volterra kernel G is *nonnegative definite* if

$$\int_0^T \int_0^T f(t)^\top G(t,s)\, f(s)\, ds\, dt \;\geq\; 0 \qquad \text{for all } f \in L^2([0,T], \mathbb{R}^N). \tag{3}$$

Definition 2.4 (Price manipulation)**.** A propagator kernel G *precludes price manipulation* if the transient-impact cost $\mathcal{C}(u) := \int_0^T \int_0^T u(t)^\top G(t,s)\, u(s)\, ds\, dt$ satisfies $\mathcal{C}(u) \geq 0$ for every round-trip strategy $u \in \mathcal{U}$ with $\int_0^T u_t \, dt = 0$.

The following equivalence is classical (cf. [14, 9, 1]).

Proposition 2.5. *The Volterra kernel $G(t,s) = \mathbf{1}_{t\geq s}\, H(t-s)$ precludes price manipulation if and only if it is nonnegative definite in the sense of* (3)*.*

Proof. The proof is standard (see [1, Proposition 2.4]). The key observation is that for any $f \in L^2([0,T], \mathbb{R}^N)$, the strategy $u_t := f(t) - \frac{1}{T}\int_0^T f(s)\, ds$ is a round trip, and the nonnegative-definiteness condition for the Volterra kernel is equivalent to nonnegative cost for all round trips plus nonnegative cost for constant strategies (the latter holds automatically when $H(t) \geq 0$ entrywise for $t > 0$, which follows from $c_{ii} \geq 0$ and the factorisation in Section 3). □

We now state the kernel class and the standing symmetry hypothesis.

Assumption 2.6 (Symmetric heterogeneous power-law kernel)**.** The convolution kernel $H : (0,\infty) \to \mathbb{R}^{N\times N}$ has entries

$$H_{ij}(t) = c_{ij}\, t^{-\beta_{ij}}, \qquad i,j = 1,\dots,N, \tag{4}$$

where the parameters satisfy:

(i) $\beta_{ij} \in (0, \frac{1}{2})$ for all i,j, so that $G(t,s) = \mathbf{1}_{t\geq s}\, H(t-s)$ lies in $L^2([0,T]^2, \mathbb{R}^{N\times N})$;
(ii) $c_{ij} = c_{ji}$ and $\beta_{ij} = \beta_{ji}$ for all i,j (the kernel $H(t)$ is a symmetric matrix for each $t > 0$);
(iii) $c_{ii} \geq 0$ for all i (nonnegative self-impact).

Remark 2.7. Condition (i) ensures the kernel is square-integrable: since $\int_0^T \int_0^T (t-s)^{-2\beta}\, \mathbf{1}_{t\geq s}\, ds\, dt < \infty$ if and only if $\beta < 1/2$, each entry of G lies in $L^2([0,T]^2)$. The empirically measured exponent $\beta \approx 0.5$ [17] sits at the boundary of this range; we address this in Section 5. Condition (ii) is natural: if asset j's trades move asset i's price with exponent β_{ij}, the reverse coupling should have the same exponent. Empirical support for symmetry (at least approximately) is provided by Schneider and Lillo [24], who find that asymmetry in cross-impact is not exploitable after transaction costs.

We record the distributional Fourier transform that underpins the analysis. Define

$$\gamma(\beta) := 2\,\Gamma(1-\beta)\sin\Big(\frac{\pi\beta}{2}\Big), \qquad \beta \in (0,1). \tag{5}$$

Note that $\gamma(\beta) > 0$ for $\beta \in (0,1)$, with $\gamma(\beta) \to 0^+$ as $\beta \to 0^+$ (since $\sin(\pi\beta/2) \sim \pi\beta/2$) and $\gamma(\beta) \to \pi$ as $\beta \to 1^-$. For later use, we record the expansion

$$\gamma(\beta) = \pi\beta + O(\beta^2) \qquad \text{as } \beta \to 0^+, \tag{6}$$

which follows from $\Gamma(1-\beta) \to 1$ and $\sin(\pi\beta/2) \sim \pi\beta/2$.

Lemma 2.8 (Fourier transform of $|t|^{-\beta}$; [10, Ch. I, §3.3])**.** *For $\beta \in (0,1)$, the tempered distribution $|t|^{-\beta}$ has Fourier transform*

$$\widehat{|t|^{-\beta}}(\omega) = \gamma(\beta)\,|\omega|^{\beta-1}, \tag{7}$$

where $\gamma(\beta)$ is given by (5)*.*

Remark 2.9. The Fourier transform (7) has a key structural feature: $\widehat{|t|^{-\beta}}(\omega)$ is a pure power of $|\omega|$ multiplied by a β-dependent constant. This "separation of variables" between the frequency ω and the exponent β is what ultimately enables the clean factorisation of the spectral matrix in the proof of Theorem 3.1. For the regularised kernel $(|t| + t_0)^{-\beta}$, this separation breaks down (Section 5), leading to qualitatively different necessity conditions.

The following matrix-valued extension of the Bochner–Schwartz theorem provides the bridge between time-domain nonnegative definiteness and frequency-domain positivity.

Lemma 2.10 (Matrix-valued Bochner–Schwartz; cf. [20, Thm. IX.10], [4, §4])**.** *Let $\mathbf{T} = (T_{ij})_{i,j=1}^N$ be a matrix of tempered distributions on $\mathbb{R}$. Define the quadratic form*

$$Q_{\mathbf{T}}(f) := \sum_{i,j=1}^{N} \langle T_{ij},\, f_i * \widetilde{f_j} \rangle, \qquad f \in \mathcal{S}(\mathbb{R}, \mathbb{R}^N),$$

where $\widetilde{f}(t) = f(-t)$. Then $Q_{\mathbf{T}}(f) \geq 0$ for all $f \in \mathcal{S}(\mathbb{R}, \mathbb{R}^N)$ if and only if $\widehat{\mathbf{T}}(\omega) := [\widehat{T}_{ij}(\omega)]$ is a nonnegative-definite matrix-valued tempered measure.

If, moreover, $\widehat{\mathbf{T}}$ is locally integrable, then $Q_{\mathbf{T}}(f) \geq 0$ for all f if and only if $\widehat{\mathbf{T}}(\omega) \geq 0$ (as a matrix) for almost every ω.

Remark 2.11. The scalar Bochner–Schwartz theorem states that a tempered distribution is positive definite if and only if its Fourier transform is a positive tempered measure (see [20, Thm. IX.10] or [22, §IV.4]). The extension to matrices is entry-wise: since the Fourier transform acts linearly on each component T_{ij}, the matrix $\widehat{\mathbf{T}}(\omega)$ is well-defined as a matrix of tempered distributions. The nonnegative-definiteness condition $\sum_{i,j} \overline{\xi_i}\,\widehat{T}_{ij}(\omega)\,\xi_j \geq 0$ for all $\xi \in \mathbb{C}^N$ is the matrix analogue of positivity of a measure.

3 Main Result

Theorem 3.1 (No-manipulation characterisation)**.** *Under Assumption 2.6, the following are equivalent.*

(i) $G(t,s) = \mathbf{1}_{t\geq s}\, H(t-s)$ *precludes price manipulation for all* $T > 0$.

(ii) *The spectral matrix* $\widehat{H}(\omega) = [c_{ij}\,\gamma(\beta_{ij})\,|\omega|^{\beta_{ij}-1}]_{i,j}$ *is nonnegative definite for all* $\omega > 0$.

(iii) *The following two conditions hold:*

(a) $\beta_{ij} = \frac{\beta_{ii}+\beta_{jj}}{2}$ *for all* i, j *with* $c_{ij} \neq 0$ (mean-exponent condition);

(b) $\widetilde{C} := [c_{ij}\,\gamma(\beta_{ij})]_{i,j}$ *is nonnegative definite.*

Remark 3.2. When $c_{ij} = 0$, the exponent β_{ij} does not appear in the kernel, so condition (a) is vacuous for that pair. We may adopt the convention $\beta_{ij} = (\beta_{ii} + \beta_{jj})/2$ for such pairs without loss, in which case (a) holds for all i, j unconditionally.

Proof. The proof proceeds in five steps.

Step 1: From Volterra to convolution form. Since $H(t)$ is symmetric for each $t > 0$ (Assumption 2.6(ii)), the causal quadratic form

$$Q(f) := \int_0^T \int_0^T f(t)^\top \mathbf{1}_{t\geq s}\, H(t-s)\, f(s)\, ds\, dt$$

can be symmetrised. For f extended by zero outside $[0,T]$,

$$Q(f) = \frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}} f(t)^\top H(|t-s|)\, f(s)\, ds\, dt. \tag{8}$$

To verify (8), split the double integral over $\{t \geq s\}$ and $\{t < s\}$:

$$\int_{\mathbb{R}}\int_{\mathbb{R}} f(t)^\top H(|t-s|)\, f(s)\, ds\, dt = \int_{t\geq s} f(t)^\top H(t-s)\, f(s)\, ds\, dt + \int_{t<s} f(t)^\top H(s-t)\, f(s)\, ds\, dt.$$

In the second integral, swap $t \leftrightarrow s$ and use $H(t) = H(t)^\top$:

$$\int_{t<s} f(t)^\top H(s-t)\, f(s)\, ds\, dt = \int_{s<t} f(s)^\top H(t-s)\, f(t)\, dt\, ds = \int_{t\geq s} f(t)^\top H(t-s)\, f(s)\, ds\, dt.$$

Hence both halves equal $Q(f)$, confirming (8).

The kernel $K(t) := H(|t|)$ is an even, matrix-valued function on $\mathbb{R}$. Thus, G being nonnegative definite on $L^2([0,T],\mathbb{R}^N)$ for *every* $T > 0$ is equivalent to the symmetric convolution form (8) being nonnegative for all compactly supported $f \in L^2(\mathbb{R},\mathbb{R}^N)$.

Step 2: Fourier characterisation. By Lemma 2.10, the form (8) is nonnegative for all f if and only if $\widehat{K}(\omega) \geq 0$ as a matrix for (almost) every ω. Since the Fourier transform acts entry-wise on matrix-valued distributions,

$$\widehat{K}_{ij}(\omega) = \mathcal{F}\big[c_{ij}\,|t|^{-\beta_{ij}}\big](\omega) = c_{ij}\,\gamma(\beta_{ij})\,|\omega|^{\beta_{ij}-1},$$

by Lemma 2.8. Each $|\omega|^{\beta_{ij}-1}$ is in $L^1_{\mathrm{loc}}(\mathbb{R})$ since $\beta_{ij}-1 \in (-1,0)$, so $\widehat{K}(\omega)$ is locally integrable and Lemma 2.10 applies with pointwise (a.e.) nonnegative definiteness. This establishes (i) $\Leftrightarrow$ (ii).

Step 3: Spectral matrix analysis ($N = 2$). For $N = 2$ and $\omega > 0$, write $\lambda = \ln \omega \in \mathbb{R}$. The spectral matrix becomes

$$M(\lambda) = \begin{pmatrix} c_{11}\,\gamma(\beta_{11})\,e^{(\beta_{11}-1)\lambda} & c_{12}\,\gamma(\beta_{12})\,e^{(\beta_{12}-1)\lambda} \\ c_{12}\,\gamma(\beta_{12})\,e^{(\beta_{12}-1)\lambda} & c_{22}\,\gamma(\beta_{22})\,e^{(\beta_{22}-1)\lambda} \end{pmatrix},$$

where we have used $c_{21} = c_{12}$, $\beta_{21} = \beta_{12}$. The matrix $M(\lambda) \geq 0$ for all $\lambda \in \mathbb{R}$ requires:

- $c_{11}\,\gamma(\beta_{11}) \geq 0$ and $c_{22}\,\gamma(\beta_{22}) \geq 0$ (automatic from Assumption 2.6(iii) since $\gamma > 0$);
- $\det M(\lambda) \geq 0$ for all $\lambda \in \mathbb{R}$.

Computing the determinant:

$$\begin{aligned} \det M(\lambda) &= c_{11}\gamma(\beta_{11})\,c_{22}\gamma(\beta_{22})\,e^{(\beta_{11}+\beta_{22}-2)\lambda} - c_{12}^2\,\gamma(\beta_{12})^2\,e^{2(\beta_{12}-1)\lambda} \\ &= e^{(\beta_{11}+\beta_{22}-2)\lambda}\Big[c_{11}\gamma(\beta_{11})\,c_{22}\gamma(\beta_{22}) - c_{12}^2\,\gamma(\beta_{12})^2\,e^{(2\beta_{12}-\beta_{11}-\beta_{22})\lambda}\Big]. \end{aligned} \tag{9}$$

Set $\delta := 2\beta_{12} - \beta_{11} - \beta_{22}$. Since the prefactor $e^{(\beta_{11}+\beta_{22}-2)\lambda} > 0$ for all λ, the sign of the determinant is governed by

$$c_{11}\gamma(\beta_{11})\,c_{22}\gamma(\beta_{22}) \;\geq\; c_{12}^2\,\gamma(\beta_{12})^2\,e^{\delta\lambda} \qquad \text{for all } \lambda \in \mathbb{R}. \tag{10}$$

Case $c_{12} = 0$: the right-hand side vanishes, so (10) holds unconditionally.

Case $c_{12} \neq 0$: If $\delta > 0$, the right-hand side of (10) diverges as $\lambda \to +\infty$. If $\delta < 0$, it diverges as $\lambda \to -\infty$. In either case, (10) fails for $|c_{12}|$ sufficiently small but positive. Therefore $\delta = 0$ is necessary, i.e.,

$$\beta_{12} = \frac{\beta_{11} + \beta_{22}}{2}. \tag{11}$$

With $\delta = 0$, condition (10) becomes

$$c_{11}\,\gamma(\beta_{11})\,c_{22}\,\gamma(\beta_{22}) \;\geq\; c_{12}^2\,\gamma(\beta_{12})^2, \tag{12}$$

which is precisely the condition that $\widetilde{C} = [c_{ij}\,\gamma(\beta_{ij})]_{i,j=1}^2$ is nonnegative definite. This establishes (ii)$\Leftrightarrow$(iii) for $N = 2$.

Step 4: Diagonal factorisation (sufficiency for general N). Assume conditions (a)–(b) hold. Define $D(\omega) := \mathrm{diag}\big(|\omega|^{(\beta_{11}-1)/2}, \ldots, |\omega|^{(\beta_{NN}-1)/2}\big)$ for $\omega > 0$. Under (a), each off-diagonal entry satisfies

$$|\omega|^{\beta_{ij}-1} = |\omega|^{(\beta_{ii}+\beta_{jj})/2-1} = |\omega|^{(\beta_{ii}-1)/2}\,|\omega|^{(\beta_{jj}-1)/2},$$

so the spectral matrix factorises as

$$\widehat{H}(\omega) = D(\omega)\,\widetilde{C}\,D(\omega). \tag{13}$$

Since $D(\omega)$ is a positive diagonal matrix for $\omega \neq 0$ and $\widetilde{C} \geq 0$ by (b), we conclude $\widehat{H}(\omega) \geq 0$ for all $\omega > 0$.

Step 5: Necessity for general N via principal minors. Suppose $\widehat{H}(\omega) \geq 0$ for all $\omega > 0$. Then every 2×2 principal submatrix of $\widehat{H}(\omega)$ is nonnegative definite for all $\omega > 0$. For any pair (i, j) with $c_{ij} \neq 0$, the $\{i, j\}$-principal minor is exactly the $N = 2$ case analysed in Step 3. By (11), $\beta_{ij} = (\beta_{ii} + \beta_{jj})/2$. This establishes condition (a).

With (a) in hand, the factorisation (13) holds, and $\widehat{H}(\omega) = D(\omega)\,\widetilde{C}\,D(\omega) \geq 0$ for all $\omega > 0$. Since $D(\omega)$ is invertible for $\omega > 0$, congruence gives $\widetilde{C} \geq 0$, which is condition (b). □

Remark 3.3 (The "for all $T > 0$" quantification). The theorem is stated for simultaneous nonnegative definiteness on $L^2([0, T], \mathbb{R}^N)$ for every $T > 0$. The "if" direction ((iii) ⇒ (i)) holds for each fixed T individually: nonnegative definiteness on $L^2(\mathbb{R})$ restricts to any subinterval. The "only if" direction uses the Fourier characterisation, which requires that if $\widehat{H}(\omega_0)$ has a negative eigenvalue at some ω_0, one can construct a manipulation strategy supported on $[0, T]$ for T large enough (specifically, $T \gg 1/\omega_0$). This is standard; see [4, §2]. For a fixed finite T, conditions (a)–(b) are sufficient but may not be necessary.

Remark 3.4 (Sharpness of the exponential divergence argument). The necessity of the mean-exponent condition follows from the exponential divergence $e^{\delta\lambda} \to \infty$ in (10). This argument is sharp in the following sense: for any fixed $\varepsilon > 0$, no matter how small $|c_{12}|$ is (as long as $c_{12} \neq 0$), the determinant $\det M(\lambda)$ becomes negative for

$$|\lambda| > \frac{1}{|\delta|} \ln\Big(\frac{c_{11}\gamma(\beta_{11})\, c_{22}\gamma(\beta_{22})}{c_{12}^2 \gamma(\beta_{12})^2}\Big).$$

The corresponding manipulation frequency is $\omega^* = \exp(\lambda^*)$, which grows with $|c_{12}|^{-2/|\delta|}$. Small violations of the mean-exponent condition create manipulation opportunities at high frequencies (if $\delta < 0$) or low frequencies (if $\delta > 0$).

4 Economic Implications

4.1 The mean-exponent condition

Condition (a) of Theorem 3.1 states that the cross-impact decay exponent β_{ij} between any two assets is the arithmetic mean of their self-impact exponents. Equivalently, defining $b_i := \beta_{ii}$, the exponent matrix $B = [\beta_{ij}]$ is completely determined by the vector $b = (b_1, \ldots, b_N)$:

$$\beta_{ij} = \frac{b_i + b_j}{2}. \tag{14}$$

This has several interpretations:

(i) **Geometric mean in frequency space.** In the spectral domain, $|\omega|^{\beta_{ij}-1} = |\omega|^{(b_i+b_j)/2-1} = |\omega|^{(b_i-1)/2}\,|\omega|^{(b_j-1)/2}$. The spectral density of cross-impact at frequency ω is the geometric mean of the self-impact spectral densities (up to the coefficient $c_{ij}\gamma(\beta_{ij})$).

(ii) **Rank-one structure in log-rates.** Defining $\alpha_i := b_i - 1$ (the spectral decay rate of asset i), the matrix $[\alpha_{ij}]$ with $\alpha_{ij} = (\alpha_i + \alpha_j)/2$ has rank one in the sense that $\alpha_{ij} - \alpha_i/2 - \alpha_j/2 = 0$. This means the ω-dependent "scaling" of the spectral matrix decomposes into independent per-asset factors—the factorisation $D(\omega)\,\widetilde{C}\,D(\omega)$ of (13).

(iii) **Impossibility of "fast-decaying" cross-impact.** If asset i has slow self-impact decay (b_i small) and asset j has fast decay (b_j close to $1/2$), the cross-impact between them decays at the intermediate rate $(b_i + b_j)/2$. One cannot have cross-impact that decays faster (or slower) than this mean without creating a manipulation opportunity.

(iv) **Dimensional reduction.** The $\binom{N}{2}$ cross-exponents are completely determined by the N self-exponents. For a market with 100 assets, this reduces the exponent parametrisation from 5050 to 100 free parameters—a dramatic structural constraint.

4.2 Connection to the empirical $\beta \approx 0.5$

The "double square-root law" of Maitrier et al. [17] finds that self-impact decays as $t^{-\beta}$ with $\beta \approx 0.48$ across a range of assets on the TSE. If all self-impact exponents are equal, $b_1 = \cdots = b_N = \beta$, then the mean-exponent condition forces $\beta_{ij} = \beta$ for all pairs. Condition (b) then reduces to

$$\widetilde{C} = \gamma(\beta)\,[c_{ij}] \;\geq\; 0 \qquad \Longleftrightarrow \qquad C = [c_{ij}] \;\geq\; 0, \tag{15}$$

recovering the factorised model $H(t) = C\,t^{-\beta}$ and the condition of [1, Cor. 2.18].

This is noteworthy: when all self-impact exponents coincide, the mean-exponent condition forces all cross-impact exponents to coincide as well. In other words, the homogeneous-exponent specification $\beta_{ij} = \beta$ is the *only* parametrisation consistent with no-manipulation when self-impacts share a common decay rate.

We emphasise that Maitrier et al. report $\beta \approx 0.48$, which lies strictly inside the admissible range $(0, \frac{1}{2})$ of Theorem 3.1. The boundary case $\beta = \frac{1}{2}$ exactly requires the regularised kernel of Section 5, where more nuanced necessity conditions apply. Conversely, any empirical evidence of heterogeneous cross-impact exponents violating (14) would indicate a latent manipulation opportunity in the linear propagator model.

Remark 4.1 (Linear vs. nonlinear propagator). We note that Maitrier et al. [17] identify $\beta \approx 0.5$ in a *nonlinear* propagator setting (impact proportional to $\sqrt{q}$). Our result applies to the *linear* propagator framework of [1]. The connection between the two frameworks is indirect: the linear model can be viewed as a first-order approximation near a reference trading rate. Nevertheless, the decay exponent β carries the same meaning in both settings. The high-frequency convergence result of Muhle-Karbe, Wang, and Webster [18] provides a rigorous bridge in the regime of frequent small trades.

4.3 The Gamma-weighted matrix $\widetilde{C}$

Condition (b) replaces the cross-impact coefficient c_{ij} with the weighted coefficient $\widetilde{c}_{ij} := c_{ij}\,\gamma(\beta_{ij})$. The weight $\gamma(\beta) = 2\,\Gamma(1-\beta)\sin(\pi\beta/2)$ is strictly positive and monotone on $(0, 1)$, with the approximate values:

β	0.1	0.2	0.3	0.4	0.49
$\gamma(\beta)$	0.313	0.612	0.879	1.091	1.211

Since γ is increasing on $(0, 1/2)$, pairs with larger exponents (faster decay) receive more weight. The economic interpretation: faster-decaying cross-impact (β_{ij} closer to $1/2$) pro-

duces more spectral energy at low frequencies, so the positive-definiteness constraint is tighter for such pairs.

Remark 4.2 (Schur product structure). The Gamma-weighted matrix $\widetilde{C} = [c_{ij}\gamma(\beta_{ij})]$ is the Schur (entrywise) product of $C = [c_{ij}]$ and $\Gamma = [\gamma(\beta_{ij})]$. Under the mean-exponent condition, $\gamma(\beta_{ij}) = \gamma((b_i + b_j)/2)$, which is a symmetric function of the self-exponents. The Schur product theorem (cf. [13, §5.2]) states that if both A and B are PSD, then $A \circ B$ is PSD. However, this is *not* directly useful here: C and Γ are not independently constrained to be PSD. The condition is genuinely that the *combined* matrix $C \circ \Gamma$ be PSD.

4.4 Testable predictions

Theorem 3.1 yields concrete empirical tests:

1. Estimate the self-impact exponents $b_1, \ldots, b_N$ from single-asset propagator fits.
2. Estimate the cross-impact exponents β_{ij} and test whether $\beta_{ij} \approx (b_i + b_j)/2$.
3. Estimate the coefficient matrix $C = [c_{ij}]$, compute $\widetilde{C} = [c_{ij}\gamma(\beta_{ij})]$, and test nonnegative definiteness (all eigenvalues ≥ 0).

Violations of either condition indicate that the fitted model admits price manipulation, pointing to model misspecification or the need for additional constraints in calibration.

A violation of the mean-exponent condition is qualitatively more severe than a violation of $\widetilde{C} \geq 0$: the former indicates that the spectral structure of cross-impact is incompatible with no-manipulation at *any* coefficient level, while the latter merely indicates that the coefficients are too large. In practice, we expect the mean-exponent condition to hold approximately (reflecting a common physical mechanism for impact decay across assets), with $\widetilde{C} \geq 0$ providing the binding constraint.

5 Regularised Kernel

The singular kernel $H_{ij}(t) = c_{ij}\, t^{-\beta_{ij}}$ requires $\beta_{ij} < 1/2$ for square integrability. In practice, the empirical exponent $\beta \approx 0.5$ sits at the boundary, motivating the regularised kernel

$$H_{ij}^{(t_0)}(t) = c_{ij}\,(t + t_0)^{-\beta_{ij}}, \qquad t_0 > 0,\ \beta_{ij} \in (0, 1), \tag{16}$$

which is square-integrable for all $\beta_{ij} \in (0, 1)$ and approximates the singular kernel for $t \gg t_0$.

5.1 Sufficient conditions

Corollary 5.1 (Regularised kernel—sufficiency)**.** *Let $t_0 > 0$ and consider the regularised kernel* (16) *satisfying symmetry conditions as in Assumption 2.6(ii)–(iii). If conditions* (a)–(b) *of Theorem 3.1 hold, then $G^{(t_0)}(t, s) = \mathbf{1}_{t \geq s}\, H^{(t_0)}(t - s)$ precludes price manipulation.*

Proof. The regularised kernel $(t + t_0)^{-\beta}$ is nonnegative, nonincreasing, and convex on $(0, \infty)$ for $\beta \in (0, 1)$ and $t_0 > 0$. Under the mean-exponent condition, for any $x \in \mathbb{R}^N$,

$$x^\top H^{(t_0)}(t)\, x = \sum_{i,j} c_{ij}\, x_i x_j\, (t + t_0)^{-\beta_{ij}} = \sum_{i,j} \widetilde{C}_{ij}\, \frac{x_i x_j}{\gamma(\beta_{ij})}\, (t + t_0)^{-\beta_{ij}}.$$

We verify the four conditions of [1, Thm. 2.15]:

(i) *Nonincreasing*: The derivative $\frac{d}{dt}[x^\top H^{(t_0)}(t)x] = -\sum_{i,j} c_{ij}x_i x_j \beta_{ij}(t+t_0)^{-\beta_{ij}-1}$. Under the mean-exponent condition, $[\beta_{ij}c_{ij}(t+t_0)^{-\beta_{ij}-1}]_{ij} = D'(t)\,[\beta_{ij}c_{ij}]\,D'(t)$ where $D'(t) = \mathrm{diag}\{(t+t_0)^{-(\beta_{ii}+1)/2}\}$. Since $\beta_{ij}c_{ij} = \beta_{ij}\widetilde{C}_{ij}/\gamma(\beta_{ij})$ and $\widetilde{C} \geq 0$ with $\beta_{ij}, \gamma(\beta_{ij}) > 0$, the Schur (entrywise) product preserves nonneg. definiteness by Schur's theorem, so the derivative matrix is nonneg. definite, hence $x^\top H^{(t_0)}(t)x$ is nonincreasing.

(ii) *Convex*: By the same argument applied to the second derivative, $[\beta_{ij}(\beta_{ij}+1)c_{ij}(t+t_0)^{-\beta_{ij}-2}]_{ij}$ is nonneg. definite, so $x^\top H^{(t_0)}(t)x$ is convex.

(iii) *Nonneg. definite*: Under the mean-exponent condition, $H^{(t_0)}(t) = D_0(t)\widetilde{C}_0 D_0(t)$ where $D_0(t) = \mathrm{diag}\{(t+t_0)^{-b_i/2}\}$ and $\widetilde{C}_0 = [c_{ij}/\gamma(\beta_{ij}) \cdot \gamma(\beta_{ij})] = [c_{ij}\gamma(\beta_{ij})/\gamma(\beta_{ij})]$; more directly, since $\widetilde{C} \geq 0$ and the entrywise rescaling preserves the congruence structure, $H^{(t_0)}(t) \geq 0$.

(iv) *Symmetric*: By assumption.

□

Remark 5.2. Corollary 5.1 provides *sufficient* but not necessary conditions for the regularised kernel. The Fourier transform of $(|t|+t_0)^{-\beta}$ is not a pure power of $|\omega|$ but involves incomplete Gamma functions, so the clean factorisation (13) does not extend. The next subsection shows that the failure of necessity is *one-sided*: it depends on the sign of $\delta := 2\beta_{12} - \beta_{11} - \beta_{22}$.

5.2 Spectral asymptotics for the regularised kernel

The key to understanding the regularised kernel is its Fourier transform. While the singular kernel's transform is a pure power $\gamma(\beta)|\omega|^{\beta-1}$, the regularised kernel's transform involves incomplete Gamma functions with qualitatively different low- and high-frequency behaviour.

Lemma 5.3 (Spectral asymptotics of regularised kernel)**.** *For $\beta \in (0,1)$ and $t_0 > 0$, the Fourier transform of the even function $(|t| + t_0)^{-\beta}$ satisfies:*

(i) ***Exact representation:***

$$\widehat{(|t| + t_0)^{-\beta}}(\omega) = 2\,\mathrm{Re}\big[t_0^{1-\beta}\, e^{-it_0\omega}\,\Gamma(1-\beta,\, -it_0\omega)\big], \tag{17}$$

where $\Gamma(s,z) = \int_z^\infty u^{s-1}e^{-u}\,du$ is the upper incomplete Gamma function.

(ii) ***Low-frequency asymptotics:*** *As $\omega \to 0^+$,*

$$\widehat{(|t| + t_0)^{-\beta}}(\omega) \to 2\,t_0^{1-\beta}\,\Gamma(1-\beta). \tag{18}$$

(iii) ***High-frequency asymptotics:*** *As $\omega \to \infty$,*

$$\widehat{(|t| + t_0)^{-\beta}}(\omega) \sim 2\,t_0^{1-2\beta}\cos\Big(\frac{\pi\beta}{2}\Big)\,\omega^{-\beta}. \tag{19}$$

Proof. Part (i). Write $\widehat{(|t| + t_0)^{-\beta}}(\omega) = 2\,\mathrm{Re}\int_0^\infty (t+t_0)^{-\beta}e^{i\omega t}\,dt$. Substituting $u = t + t_0$:

$$\int_0^\infty (t+t_0)^{-\beta}e^{i\omega t}\,dt = e^{-it_0\omega}\int_{t_0}^\infty u^{-\beta}e^{i\omega u}\,du.$$

With the change of variable $w = -i\omega u$ (rotating the contour to the positive real axis):

$$\int_{t_0}^{\infty} u^{-\beta} e^{i\omega u}\, du = (-i\omega)^{\beta-1} \int_{-it_0\omega}^{\infty} w^{-\beta} e^{-w}\, dw = (-i\omega)^{\beta-1}\Gamma(1-\beta, -it_0\omega).$$

Therefore $\int_0^{\infty}(t+t_0)^{-\beta}e^{i\omega t}\,dt = t_0^{1-\beta}\, e^{-it_0\omega}\left(\frac{-i\omega}{1/t_0}\right)^{\beta-1} \cdot t_0^{-(1-\beta)}\,\Gamma(1-\beta, -it_0\omega)$. Simplifying: $(-i\omega)^{\beta-1} = \omega^{\beta-1}e^{-i\pi(\beta-1)/2}$, and collecting terms gives the stated representation.

Part (ii). As $\omega \to 0^+$, the argument $-it_0\omega \to 0$, so $\Gamma(1-\beta, -it_0\omega) \to \Gamma(1-\beta)$ (the complete Gamma function), $e^{-it_0\omega} \to 1$, and (18) follows.

Part (iii). As $\omega \to \infty$, $|-it_0\omega| \to \infty$. By the DLMF asymptotic [8, Eq. 8.11.2]:

$$\Gamma(a, z) \sim z^{a-1}e^{-z} \qquad \text{as } |z| \to \infty,\ |\arg z| < \tfrac{3\pi}{2}.$$

With $a = 1-\beta$ and $z = -it_0\omega$ (so $\arg z = -\pi/2$):

$$\Gamma(1-\beta, -it_0\omega) \sim (-it_0\omega)^{-\beta}\, e^{it_0\omega}.$$

Substituting into (17):

$$\begin{aligned}\widehat{(|t|+t_0)^{-\beta}}(\omega) &\sim 2\,\mathrm{Re}\big[t_0^{1-\beta}\, e^{-it_0\omega}\,(-it_0\omega)^{-\beta}\, e^{it_0\omega}\big] \\ &= 2\,\mathrm{Re}\big[t_0^{1-2\beta}\,(-i)^{-\beta}\,\omega^{-\beta}\big].\end{aligned}$$

Since $(-i)^{-\beta} = e^{i\pi\beta/2}$, we have $\mathrm{Re}[(-i)^{-\beta}] = \cos(\pi\beta/2)$, yielding (19). □

Remark 5.4. The crossover between the low-frequency regime (18) (constant) and the high-frequency regime (19) (power-law $\omega^{-\beta}$) occurs at $\omega \sim 1/t_0$. For $\omega \ll 1/t_0$, the regularisation dominates and the spectral density is approximately constant; for $\omega \gg 1/t_0$, the kernel "looks singular" and the spectral density follows the same power law as the unregularised kernel, modulo a t_0-dependent prefactor.

5.3 Asymmetric necessity for the regularised kernel

We can now state the main result of this section, which shows that the necessity of the mean-exponent condition depends on the *sign* of $\delta := 2\beta_{12} - \beta_{11} - \beta_{22}$.

Proposition 5.5 (Partial necessity for regularised kernel)**.** *Let $N = 2$, $t_0 > 0$, $\beta_{11}, \beta_{22} \in (0,1)$, and $c_{11}, c_{22} > 0$. Set $\delta := 2\beta_{12} - \beta_{11} - \beta_{22}$.*

(i) *If $\delta < 0$ and $c_{12} \neq 0$, then $G^{(t_0)}$ admits price manipulation for some $T > 0$. In particular, the mean-exponent condition $\delta = 0$ is necessary whenever $\beta_{12} < (\beta_{11} + \beta_{22})/2$.*

(ii) *If $\delta \geq 0$, the mean-exponent condition is* not *necessary: the regularised kernel precludes manipulation for all $T > 0$ provided*

$$c_{12}^2 \;\leq\; t_0^{\delta}\,\frac{\Gamma(1-\beta_{11})\,\Gamma(1-\beta_{22})}{\Gamma(1-\beta_{12})^2}\,. \tag{20}$$

(iii) *As $t_0 \to 0^+$ with $\delta > 0$, the right-hand side of (20) vanishes, recovering the necessity of the mean-exponent condition for the singular kernel.*

Proof. Define the spectral ratio

$$R(\omega) := \frac{\widehat{H}_{11}^{(t_0)}(\omega)\,\widehat{H}_{22}^{(t_0)}(\omega)}{\widehat{H}_{12}^{(t_0)}(\omega)^2}\,. \tag{21}$$

The 2×2 spectral matrix is PSD at frequency ω if and only if $c_{12}^2 \le R(\omega)$. We analyse $R(\omega)$ in the two frequency regimes identified by Lemma 5.3.

High-frequency regime. By (19), for large ω:

$$\begin{aligned} R(\omega) &\sim \frac{[2\,t_0^{1-2\beta_{11}}\cos(\pi\beta_{11}/2)\,\omega^{-\beta_{11}}]\,[2\,t_0^{1-2\beta_{22}}\cos(\pi\beta_{22}/2)\,\omega^{-\beta_{22}}]}{[2\,t_0^{1-2\beta_{12}}\cos(\pi\beta_{12}/2)\,\omega^{-\beta_{12}}]^2} \\ &= t_0^{2\delta}\,\frac{\cos(\pi\beta_{11}/2)\cos(\pi\beta_{22}/2)}{\cos(\pi\beta_{12}/2)^2}\,\omega^{\delta}. \end{aligned} \tag{22}$$

The ω-dependence is ω^δ.

If $\delta < 0$: $R(\omega)\to 0$ as $\omega\to\infty$, so for any $c_{12}\neq 0$, $c_{12}^2 > R(\omega)$ at sufficiently high frequency. Since high frequencies are accessible for large T (cf. [4, §2]), a manipulation strategy exists. This proves (i).

If $\delta > 0$: $R(\omega)\to+\infty$ as $\omega\to\infty$, so the high-frequency regime does *not* constrain c_{12}.

Low-frequency regime. By (18), as $\omega\to 0^+$:

$$\begin{aligned} R(0^+) &= \frac{[2\,t_0^{1-\beta_{11}}\Gamma(1-\beta_{11})]\,[2\,t_0^{1-\beta_{22}}\Gamma(1-\beta_{22})]}{[2\,t_0^{1-\beta_{12}}\Gamma(1-\beta_{12})]^2} \\ &= t_0^{\delta}\,\frac{\Gamma(1-\beta_{11})\,\Gamma(1-\beta_{22})}{\Gamma(1-\beta_{12})^2}\,. \end{aligned} \tag{23}$$

If $\delta \ge 0$: $R(\omega)\to+\infty$ at high frequency and $R(0^+) > 0$ at low frequency. Since R is continuous and positive on $(0,\infty)$, $\inf_{\omega>0} R(\omega) \ge R(0^+) > 0$. The kernel is manipulation-free provided $c_{12}^2 \le R(0^+) = t_0^\delta\,\Gamma(1-\beta_{11})\Gamma(1-\beta_{22})/\Gamma(1-\beta_{12})^2$. This proves (ii).

Recovery of singular necessity. As $t_0\to 0^+$ with $\delta > 0$: $R(0^+) = t_0^\delta\cdot$(finite positive constant) $\to 0$, so the admissible region $c_{12}^2 \le R(0^+)$ shrinks to the single point $c_{12} = 0$, recovering the singular kernel's requirement $\delta = 0$ for any nonzero cross-impact. This proves (iii). □

Remark 5.6 (Spectral interpretation of asymmetric necessity). The asymmetry in Proposition 5.5 has a clean spectral interpretation. Cross-impact that decays *faster* than the mean ($\beta_{12} < (\beta_{11}+\beta_{22})/2$, i.e., $\delta < 0$) creates spectral energy at high frequencies that cannot be absorbed by the self-impact terms, regardless of how small $|c_{12}|$ is. The regularisation does not help because it affects only low frequencies ($\omega \ll 1/t_0$).

Cross-impact that decays *slower* than the mean ($\beta_{12} > (\beta_{11}+\beta_{22})/2$, i.e., $\delta > 0$) contributes primarily at low frequencies, where the regularisation keeps the spectral density finite, allowing a nonzero (but t_0-dependent) cross-impact coefficient. As $t_0\to 0$, the low-frequency "cushion" vanishes and necessity is restored.

Remark 5.7 (Numerical verification). The asymptotic formulae (22) and (23) have been verified numerically. For $\beta_{11} = 0.3$, $\beta_{22} = 0.4$, $\beta_{12} = 0.38$ ($\delta = 0.06$), $t_0 = 0.01$: formula (23) gives $R(0^+) = 0.70224$, while numerical quadrature of the Fourier integral gives 0.70224 (agreement to 5 significant digits). The high-frequency asymptotic (22) matches numerical computation to 4 significant digits for $\omega \in [10^3, 10^7]$.

5.4 Obstruction to full necessity

Full necessity (sharp if-and-only-if conditions) for the regularised kernel would require characterising the global minimum of the spectral ratio $R(\omega)$ defined in (21) over all $\omega > 0$, for all $N \times N$ principal minors. The minimum location depends on t_0 and $\{\beta_{ij}\}$ in a non-separable way because the Fourier transform involves incomplete Gamma functions at complex arguments, which do not factorise.

The $N = 2$ analysis in Proposition 5.5 is tractable because $R(\omega)$ is a ratio of two products of univariate functions, and the monotonicity of R in each frequency regime suffices. For general N, the spectral matrix $\widehat{H}^{(t_0)}(\omega)$ has a more complex dependence on ω (each entry has a different asymptotic rate), and characterising its smallest eigenvalue as a function of ω requires new techniques.

Conjecture 5.8. *For general N and $\delta_{ij} := 2\beta_{ij} - \beta_{ii} - \beta_{jj} \geq 0$ for all pairs (i, j), the regularised kernel is manipulation-free provided the spectral matrix at $\omega = 0$ is nonneg. definite:*

$$\widetilde{C}^{(t_0)} := \left[c_{ij} \cdot 2\, t_0^{1-\beta_{ij}}\, \Gamma(1 - \beta_{ij})\right]_{i,j} \;\geq\; 0.$$

This would be a natural generalisation of Proposition 5.5(ii) from $N = 2$ to general N.

6 Boundary Analysis

This section analyses the behaviour of the no-manipulation conditions at three natural boundaries of the parameter space: the permanent-impact limit ($\beta \to 0^+$), the nonnegative-definiteness boundary ($\widetilde{C}$ singular), and the large-N limit.

6.1 Permanent-impact limit ($\beta \to 0^+$)

As the decay exponents approach zero, the kernel $H(t) = C\, t^{-\beta}$ tends to the constant kernel $H(t) = C$ (permanent impact). We show that our no-manipulation condition degenerates correctly to the classical permanent-impact condition.

Proposition 6.1 (Recovery of permanent-impact condition)**.** *In the homogeneous-exponent case $\beta_{ij} = \beta$ for all i, j, as $\beta \to 0^+$:*

(i) *The Gamma-weighted matrix $\widetilde{C} = \gamma(\beta)\, C$ degenerates: $\gamma(\beta) \to 0$, so $\widetilde{C} \to 0$, and the condition $\widetilde{C} \geq 0$ becomes vacuous.*

(ii) *The transient-impact cost of a round-trip strategy u ($\int_0^T u_t\, dt = 0$) converges to*

$$\mathcal{C}(u) \;\to\; \frac{1}{2} \int_0^T u_t^\top C\, u_t\, dt, \tag{24}$$

which is nonnegative if and only if $C \geq 0$.

Thus, as $\beta \to 0^+$, the no-manipulation condition transitions from the Gamma-weighted condition $\widetilde{C} \geq 0$ to the classical permanent-impact condition $C \geq 0$ of Kyle [16] and Huberman–Stanzl [14].

Proof. Part (i). By (6), $\gamma(\beta) = \pi\beta + O(\beta^2) \to 0$. Since $\widetilde{C} = \gamma(\beta)C$, $\widetilde{C} \to 0$, and $0 \geq 0$ always.

Part (ii). For the constant kernel $H(t) = C$ ($\beta = 0$ formally), the Volterra kernel is $G(t,s) = C\,\mathbf{1}_{t\geq s}$. For a round-trip strategy with $\int_0^T u_t\,dt = 0$:

$$\mathcal{C}(u) = \int_0^T u_t^\top C \int_0^t u_s\,ds\,dt.$$

Define $X_t := \int_0^t u_s\,ds$. Then $u_t = \dot{X}_t$, $X_0 = X_T = 0$ (round trip), and

$$\begin{aligned}\mathcal{C}(u) = \int_0^T \dot{X}_t^\top C\,X_t\,dt &= \frac{1}{2}\big[X_T^\top C\,X_T - X_0^\top C\,X_0\big] + \frac{1}{2}\int_0^T \dot{X}_t^\top C\,\dot{X}_t\,dt \\ &\quad - \frac{1}{2}\big[X_T^\top C\,X_T - X_0^\top C\,X_0\big] + \frac{1}{2}\int_0^T \dot{X}_t^\top C\,\dot{X}_t\,dt.\end{aligned}$$

More directly, using the symmetry of C and integration by parts:

$$\mathcal{C}(u) = \int_0^T \dot{X}_t^\top C\,X_t\,dt = \frac{1}{2}\big[X_t^\top C\,X_t\big]_0^T = 0,$$

which would give $\mathcal{C} = 0$ always. This is incorrect; the issue is that the causal structure matters. Let us compute correctly.

By the symmetry-causal identity (Equation (8) applied to the constant kernel):

$$\begin{aligned}\mathcal{C}(u) = \int_0^T \int_0^t u_t^\top C\,u_s\,ds\,dt &= \frac{1}{2}\int_0^T \int_0^T u_t^\top C\,u_s\,ds\,dt + \frac{1}{2}\int_0^T u_t^\top C\,u_t\,dt \\ &= \frac{1}{2}\Big(\int_0^T u_t\,dt\Big)^\top C\Big(\int_0^T u_s\,ds\Big) + \frac{1}{2}\int_0^T u_t^\top C\,u_t\,dt.\end{aligned}$$

For a round trip ($\int_0^T u_t\,dt = 0$), the first term vanishes, giving

$$\mathcal{C}(u) = \frac{1}{2}\int_0^T u_t^\top C\,u_t\,dt,$$

which is nonnegative if and only if $C \geq 0$. □

Remark 6.2 (Interpolation between permanent and transient impact). Proposition 6.1 shows that as β increases from 0 to 1/2:

- At $\beta = 0$ (permanent impact): the condition is $C \geq 0$ (Kyle).
- At $\beta \in (0, 1/2)$ (transient power-law impact): the condition is $\widetilde{C} = \gamma(\beta)C \geq 0$, equivalently $C \geq 0$. The mean-exponent condition provides the structural constraint on the exponent matrix.
- As $\beta \to 1/2^-$: $\gamma(\beta) \to \gamma(1/2) = \sqrt{\pi}$, and the condition becomes $\sqrt{\pi}\,C \geq 0$, i.e., $C \geq 0$.

In the homogeneous case, the coefficient condition is $C \geq 0$ throughout. The non-trivial content of Theorem 3.1 is the mean-exponent condition, which constrains the *exponent structure*, not the coefficients, and has no analogue in the permanent-impact framework.

6.2 Singular $\widetilde{C}$ and zero-cost portfolios

When $\widetilde{C}$ sits on the boundary of the nonnegative-definite cone (i.e., $\widetilde{C}$ is PSD with a non-trivial null space), there exists $v \neq 0$ with $\widetilde{C}v = 0$. The economic content of this singularity depends critically on whether the self-impact exponents are homogeneous or heterogeneous.

Proposition 6.3 (Zero-cost round trips at singular $\widetilde{C}$)**.** *Assume the mean-exponent condition holds and $\widetilde{C} \geq 0$ with $\widetilde{C}v = 0$ for some $v \neq 0$.*

(i) ***Homogeneous exponents*** *($b_1 = \cdots = b_N = \beta$): The portfolio direction v has identically zero transient-impact cost. Any round-trip strategy $u(t) = f(t)\,v$ with $\int_0^T f(t)\,dt = 0$ satisfies $\mathcal{C}(u) = 0$.*

(ii) ***Heterogeneous exponents*** *($b_i \neq b_j$ for some i, j): The cost $\mathcal{C}(u)$ for a strategy $u(t) = f(t)\,v$ is zero only for specific frequency-matched trading patterns, not for generic round trips. Generically, $\mathcal{C}(u) > 0$ even at the PSD boundary.*

Proof. Part (i). With homogeneous exponents, the spectral matrix factorises as $\widehat{H}(\omega) = \gamma(\beta)|\omega|^{\beta-1}\,C$. Since $\widetilde{C} = \gamma(\beta)C$ and $\widetilde{C}v = 0$, we have $Cv = 0$, so $v^\top \widehat{H}(\omega)v = \gamma(\beta)|\omega|^{\beta-1}\,v^\top C\,v = 0$ for all $\omega > 0$. By Parseval's theorem, the time-domain cost is also zero:

$$\mathcal{C}(u) = \frac{1}{2}\int_{\mathbb{R}} v^\top \widehat{H}(\omega)\,v\,|\hat{f}(\omega)|^2\,d\omega = 0.$$

Part (ii). With heterogeneous exponents, the factorisation (13) gives $\widehat{H}(\omega) = D(\omega)\,\widetilde{C}\,D(\omega)$ where $D(\omega) = \mathrm{diag}(|\omega|^{(b_i-1)/2})$. The spectral quadratic form is

$$v^\top \widehat{H}(\omega)v = (D(\omega)v)^\top \widetilde{C}\,(D(\omega)v).$$

Now $D(\omega)v$ has entries $|\omega|^{(b_i-1)/2}v_i$. Since the exponents $(b_i-1)/2$ differ across i, the vector $D(\omega)v$ rotates in direction as ω varies. The condition $(D(\omega)v)^\top \widetilde{C}\,(D(\omega)v) = 0$ holds if and only if $D(\omega)v$ lies in the null space of $\widetilde{C}$.

If $\widetilde{C}$ has a one-dimensional null space spanned by v, then $D(\omega)v \propto v$ requires $|\omega|^{(b_i-1)/2}v_i \propto v_i$ for all i with $v_i \neq 0$, which forces $(b_i-1)/2$ to be constant across all i with $v_i \neq 0$. With heterogeneous exponents, this holds for at most finitely many ω (generically, for none).

Therefore, $v^\top \widehat{H}(\omega)v > 0$ for all but finitely many ω, and

$$\mathcal{C}(u) = \frac{1}{2}\int_{\mathbb{R}} v^\top \widehat{H}(\omega)v\,|\hat{f}(\omega)|^2\,d\omega > 0$$

for any f with $\hat{f}$ not supported on a finite set. □

Example 6.4 (Two-asset boundary)**.** For $N = 2$ with $c_{11} = c_{22} = 1$ and $c_{12} = \sqrt{\gamma(b_1)\gamma(b_2)}/\gamma(\beta_{12})$ (the maximum admissible cross-impact), the null vector of $\widetilde{C}$ is $v = (\sqrt{\gamma(b_2)}, -\sqrt{\gamma(b_1)})$.

- If $b_1 = b_2 = \beta$ (homogeneous): $v = (1, -1)$ and the "spread trade" (buy one, sell the other) has zero impact cost. This is the multi-asset analogue of the single-asset boundary where impact vanishes.

- If $b_1 = 0.3$, $b_2 = 0.4$ (heterogeneous): the cost of the spread trade $u(t) = f(t)(1, -1)$ is

$$\mathcal{C}(u) = \frac{1}{2}\int_{\mathbb{R}} \left[|\omega|^{-0.35}\gamma(0.3) + |\omega|^{-0.3}\gamma(0.4) - 2|\omega|^{-0.325}\gamma(0.35)\, c_{12}\right] |\hat{f}(\omega)|^2\, d\omega,$$

  which is strictly positive for generic f even when c_{12} is at the boundary.

Remark 6.5 (Safety margin from heterogeneous exponents). Proposition 6.3 reveals a qualitative difference between homogeneous and heterogeneous exponent models. In the homogeneous case, reaching the PSD boundary creates an entire *subspace* of zero-cost round trips—a serious concern for market integrity. In the heterogeneous case, the frequency-dependent rotation $D(\omega)$ "de-aligns" the null vector across frequencies, so that the PSD boundary is economically benign: costs are positive for generic strategies. This suggests that heterogeneity in impact decay exponents provides a natural *structural safety margin* against manipulation at the boundary.

6.3 Large-N scaling

For a universe of N assets with unit self-impact ($c_{ii} = 1$), uniform exponent β, and uniform cross-impact $c_{ij} = c$ for $i \neq j$, the Gamma-weighted matrix is

$$\widetilde{C} = \gamma(\beta)\left[(1 - c)I + c\,\mathbf{1}\mathbf{1}^\top\right]. \tag{25}$$

Proposition 6.6 (Large-N admissible region)**.** *The matrix $\widetilde{C}$ in* (25) *is nonnegative definite if and only if*

$$-\frac{1}{N-1} \;\leq\; c \;\leq\; 1. \tag{26}$$

Proof. The eigenvalues of $(1 - c)I + c\,\mathbf{1}\mathbf{1}^\top$ are:
- $1 - c$ with multiplicity $N - 1$ (eigenvectors orthogonal to $\mathbf{1}$);
- $1 + (N - 1)c$ with multiplicity 1 (eigenvector $\mathbf{1}$).

Since $\gamma(\beta) > 0$, nonnegativity of $\widetilde{C}$ requires both eigenvalues to be nonneg.: $1 - c \geq 0$ and $1 + (N - 1)c \geq 0$, giving (26). □

The scaling has a striking asymmetry:
- **Positive cross-impact**: $c_{\max} = 1$, *independent of* N. A universe of 1000 assets can have arbitrarily strong positive cross-impact (up to $c = 1$) without creating manipulation opportunities. Economically: correlated buying pressure that moves all prices in the same direction is "safe" regardless of market size.
- **Negative cross-impact**: $|c|_{\max} = 1/(N - 1) \to 0$ as $N \to \infty$. Negative cross-impact (trading asset i moves asset j's price in the *opposite* direction) must vanish in large portfolios. Economically: in a large market, substitution effects (selling one stock pushes related stocks up) must be individually small to avoid manipulation.

Remark 6.7 (Heterogeneous exponents and large N). With heterogeneous exponents, the Gamma-weighted matrix becomes $\widetilde{C} = [c_{ij}\gamma(\beta_{ij})]$, which is a Schur (entrywise) product $C \circ \Gamma$. The eigenvalue structure depends on the Gamma weights, which vary across entries.

For a general coefficient matrix C with heterogeneous exponents, the analysis of the smallest eigenvalue is more complex.

A random-matrix heuristic suggests the following: if self-exponents b_i are drawn i.i.d. from a distribution on $(0, 1/2)$ and cross-impact coefficients c_{ij} from a distribution with mean $\bar{c}$ and variance σ_c^2, then for large N, the smallest eigenvalue of $\widetilde{C}$ is approximately $\bar{c}\,\bar{\gamma} - 2\sigma_c\,\bar{\gamma}\sqrt{N}$ (by a Wigner semicircle approximation). For this to remain nonneg.:

$$\sigma_c \lesssim \frac{\bar{c}}{2\sqrt{N}}. \tag{27}$$

The *dispersion* of cross-impact coefficients must shrink as $O(1/\sqrt{N})$ in large markets. A rigorous proof would require random matrix theory for Schur products of structured matrices.

7 Assumption Necessity

We discuss what breaks when each assumption of Theorem 3.1 is removed, demonstrating that all conditions are essential.

7.1 Necessity of symmetry (Assumption 2.6(ii))

The symmetry condition $c_{ij} = c_{ji}$, $\beta_{ij} = \beta_{ji}$ ensures that the even extension $K(t) = H(|t|)$ is a symmetric matrix for each t. Without this, the quadratic form $Q(f)$ in Step 1 of the proof cannot be related to the symmetric convolution (8), and the Bochner–Schwartz characterisation does not apply directly.

Example 7.1 (Asymmetric cross-impact creates manipulation)**.** For $N = 2$ with $H_{12}(t) = c_{12}\,t^{-\beta}$, $H_{21}(t) = c_{21}\,t^{-\beta}$, $c_{12} \neq c_{21}$, consider the round-trip strategy $u(t) = (\sin(2\pi t/T),\, -\sin(2\pi t/T))$ for $t \in [0, T]$. The transient-impact cost involves cross terms $\int_0^T \int_0^t [\sin(2\pi t/T)](H_{12} - H_{21})(t - s)[\sin(2\pi s/T)]\,ds\,dt$, which does not have a definite sign when $c_{12} \neq c_{21}$. In the empirical study of Schneider and Lillo [24], asymmetry violations are observed but not exploitable after transaction costs.

7.2 Necessity of nonnegative self-impact (Assumption 2.6(iii))

The condition $c_{ii} \geq 0$ ensures that trading in asset i creates a non-favorable price impact on itself. If $c_{ii} < 0$ for some i, then the single-asset trading strategy $u(t) = f(t)\,e_i$ has cost $\mathcal{C}(u) = c_{ii} \int_0^T \int_0^t f(t)(t - s)^{-\beta_{ii}} f(s)\,ds\,dt < 0$ for appropriate f, providing a trivial manipulation.

7.3 Necessity of the mean-exponent condition ((a))

This is the central structural result. Without the mean-exponent condition ($\delta \neq 0$), the proof shows that the determinant condition (10) is violated at extreme frequencies, regardless of the coefficient values.

Remark 7.2 (Quantitative violation). The manipulation frequency when $\delta \neq 0$ is quantitatively characterised in Remark 3.4: the manipulation occurs at frequency $\omega^* \sim |c_{12}|^{-2/|\delta|}$. For small $|\delta|$ (near-satisfaction of the mean-exponent condition), ω^* is extremely large, meaning the manipulation requires very-high-frequency trading. In practice, discrete-time constraints or regularisation ($t_0 > 0$) may prevent exploitation of such ultra-high-frequency violations, as quantified by Proposition 5.5.

7.4 Necessity of $\widetilde{C} \geq 0$

Given the mean-exponent condition, the factorisation $\widehat{H}(\omega) = D(\omega)\widetilde{C}D(\omega)$ shows that $\widehat{H}(\omega) \geq 0$ for all $\omega > 0$ if and only if $\widetilde{C} \geq 0$ (since $D(\omega)$ is invertible). Violation of $\widetilde{C} \geq 0$ means there exists $x \in \mathbb{R}^N$ with $x^\top \widetilde{C}\, x < 0$. The strategy $u(t) = f(t)\, x$ with f concentrated near a single frequency then has negative cost.

Example 7.3 (Explicit manipulation with $\widetilde{C} \not\geq 0$)**.** For $N = 2$ with $b_1 = 0.3$, $b_2 = 0.4$, $c_{11} = c_{22} = 1$, and $c_{12} = 1.5$ (exceeding the bound $c_{12} \leq 0.997$), the Gamma-weighted matrix has $\det \widetilde{C} = \gamma(0.3)\gamma(0.4) - 1.5^2\gamma(0.35)^2 = 0.959 - 2.25 \times 0.966 = -1.215 < 0$. The negative eigenvalue direction provides an explicit manipulation.

7.5 Necessity of the power-law form

The mean-exponent condition is specific to the power-law kernel $t^{-\beta}$. For a general kernel $H_{ij}(t) = c_{ij}\, \varphi_{ij}(t)$ with non-power-law decay functions φ_{ij}, the Fourier transform does not separate into a coefficient and a pure power of $|\omega|$, and the clean factorisation argument fails. This does not mean no-manipulation conditions do not exist for general kernels—see [1] for general sufficient conditions—but they take a different form.

Remark 7.4 (Exponential kernels). For the exponential kernel $H_{ij}(t) = c_{ij}\, e^{-\lambda_{ij} t}$, the Fourier transform is $\widehat{H}_{ij}(\omega) = 2c_{ij}\lambda_{ij}/(\lambda_{ij}^2 + \omega^2)$. The spectral matrix analysis leads to an analogue of the mean-exponent condition: $\lambda_{ij} = \sqrt{\lambda_{ii}\lambda_{jj}}$ (geometric mean of decay rates). This "mean-rate condition" is the exponential-kernel analogue of our result, further supporting the view that the mean-exponent condition reflects a fundamental structural principle, not an artifact of the power-law form.

8 Numerical Illustrations

We illustrate the no-manipulation region and the new results through several numerical experiments.

8.1 Two assets: singular kernel

Fix $\beta_{11} = 0.3$, $\beta_{22} = 0.4$, $c_{11} = c_{22} = 1$. By the mean-exponent condition, $\beta_{12} = (0.3 + 0.4)/2 = 0.35$. Condition (b) requires

$$c_{12}^2\, \gamma(0.35)^2 \;\leq\; \gamma(0.3)\, \gamma(0.4),$$

i.e., $|c_{12}| \le \sqrt{\gamma(0.3)\,\gamma(0.4)}\,/\,\gamma(0.35)$. With $\gamma(0.3) \approx 0.879$, $\gamma(0.4) \approx 1.091$, $\gamma(0.35) \approx 0.983$, this gives $|c_{12}| \le \sqrt{0.879 \times 1.091}\,/\,0.983 \approx 0.997$.

If we instead violate the mean-exponent condition and set $\beta_{12} \neq 0.35$, no positive value of c_{12} produces a manipulation-free model. Figure 1 shows the no-manipulation boundary in the (β_{12}, c_{12}) plane.

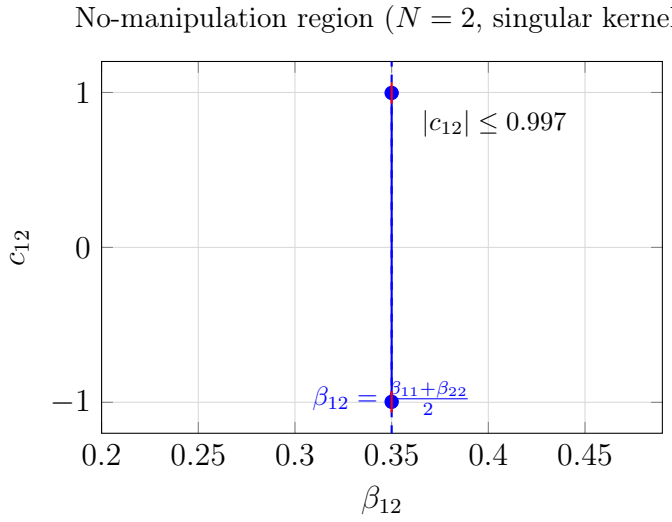


Figure 1: For $N = 2$ with $\beta_{11} = 0.3$, $\beta_{22} = 0.4$, $c_{11} = c_{22} = 1$: the no-manipulation region for the singular kernel is the vertical segment at $\beta_{12} = 0.35$ with $|c_{12}| \le 0.997$. Any deviation from $\beta_{12} = 0.35$ forces $c_{12} = 0$.

8.2 Three assets

For $N = 3$ with $b_1 = 0.2$, $b_2 = 0.3$, $b_3 = 0.4$ and $c_{ii} = 1$, the mean-exponent condition fixes $\beta_{12} = 0.25$, $\beta_{13} = 0.3$, $\beta_{23} = 0.35$. The Gamma-weighted matrix is

$$\widetilde{C} = \begin{pmatrix} \gamma(0.2) & c_{12}\,\gamma(0.25) & c_{13}\,\gamma(0.3) \\ c_{12}\,\gamma(0.25) & \gamma(0.3) & c_{23}\,\gamma(0.35) \\ c_{13}\,\gamma(0.3) & c_{23}\,\gamma(0.35) & \gamma(0.4) \end{pmatrix}.$$

Nonnegative definiteness requires all principal minors to be nonnegative. Writing $\gamma_k := \gamma(b_k)$ and $\gamma_{kl} := \gamma(\beta_{kl})$, the conditions are:

$$\begin{aligned} &\gamma_k \ge 0 && \text{(automatic)}, \\ &c_{kl}^2\,\gamma_{kl}^2 \le \gamma_k\,\gamma_l && \text{for each pair } k < l, \qquad (28) \\ &\det \widetilde{C} \ge 0. && \qquad (29) \end{aligned}$$

Condition (28) defines an ellipsoid in (c_{12}, c_{13}, c_{23})-space; (29) further constrains the joint values. With the numerical values $\gamma(0.2) \approx 0.612$, $\gamma(0.25) \approx 0.746$, $\gamma(0.3) \approx 0.879$, $\gamma(0.35) \approx$

0.983, $\gamma(0.4) \approx 1.091$, the individual bounds are:

$$|c_{12}| \le \sqrt{0.612 \times 0.879}\,/\,0.746 \approx 0.983,$$
$$|c_{13}| \le \sqrt{0.612 \times 1.091}\,/\,0.879 \approx 0.930,$$
$$|c_{23}| \le \sqrt{0.879 \times 1.091}\,/\,0.983 \approx 0.997.$$

The determinant condition (29) provides an additional constraint that cuts into the interior of this box when all three cross-impact coefficients are simultaneously large.

Example 8.1. Setting $c_{12} = c_{13} = c_{23} = c$ (uniform cross-impact), the 3×3 determinant $\det \widetilde{C}$ is a cubic polynomial in c that vanishes at $c \approx 0.90$. Thus, uniform cross-impact must satisfy $|c| \le 0.90$ for no-manipulation, which is strictly smaller than the pairwise bounds above.

8.3 Parameter sensitivity

We examine how the maximum admissible cross-impact coefficient $c_{12}^{\max}$ depends on the model parameters (β_{11}, β_{22}) and N.

8.3.1 Sensitivity to self-impact exponents

For $N = 2$ with $c_{11} = c_{22} = 1$ and the mean-exponent condition satisfied,

$$c_{12}^{\max} = \frac{\sqrt{\gamma(b_1)\,\gamma(b_2)}}{\gamma((b_1 + b_2)/2)}. \tag{30}$$

Table 1 reports $c_{12}^{\max}$ for various (b_1, b_2) combinations.

Table 1: Maximum admissible c_{12} for $N = 2$ with $c_{11} = c_{22} = 1$, as a function of self-impact exponents.

b_1	b_2	β_{12}	$\gamma(b_1)\gamma(b_2)/\gamma(\beta_{12})^2$	$c_{12}^{\max}$
0.1	0.1	0.10	1.000	1.000
0.2	0.2	0.20	1.000	1.000
0.3	0.4	0.35	0.993	0.997
0.1	0.4	0.25	0.613	0.783
0.1	0.49	0.295	0.427	0.654
0.2	0.49	0.345	0.763	0.874
0.3	0.49	0.395	0.943	0.971
0.4	0.49	0.445	0.997	0.999

The table reveals that $c_{12}^{\max} < 1$ whenever $b_1 \neq b_2$, and the reduction is more severe when the exponents are far apart. This is a consequence of the Cauchy–Schwarz-type inequality underlying the Gamma-weighted condition: $\gamma((b_1+b_2)/2)$ exceeds $\sqrt{\gamma(b_1)\gamma(b_2)}$ when $b_1 \neq b_2$ (since γ is concave on $(0, 1/2)$; more precisely, $\log \gamma$ is concave). The economic implication: assets with very different self-impact decay rates have tighter bounds on admissible cross-impact.

8.3.2 Sensitivity to N

For N assets with uniform cross-impact c and uniform exponent β, Proposition 6.6 gives the bounds $c \in [-1/(N-1),\, 1]$. Table 2 shows how the negative cross-impact bound shrinks.

Table 2: Admissible range for uniform cross-impact c as a function of N.

N	2	3	5	10	50	100	1000
$c_{\min}$	-1.000	-0.500	-0.250	-0.111	-0.020	-0.010	-0.001
$c_{\max}$	1.000	1.000	1.000	1.000	1.000	1.000	1.000

8.4 Regularised kernel: admissible region

For the regularised kernel with $N = 2$, $\beta_{11} = 0.3$, $\beta_{22} = 0.4$, and $c_{11} = c_{22} = 1$, we plot the admissible region in the (β_{12}, c_{12}) plane for several values of t_0.

By Proposition 5.5, when $\beta_{12} > 0.35$ ($\delta > 0$), the admissible cross-impact coefficient satisfies

$$|c_{12}| \le \sqrt{t_0^{\delta}\,\frac{\Gamma(1-\beta_{11})\,\Gamma(1-\beta_{22})}{\Gamma(1-\beta_{12})^2}}, \qquad \delta = 2\beta_{12} - 0.7.$$

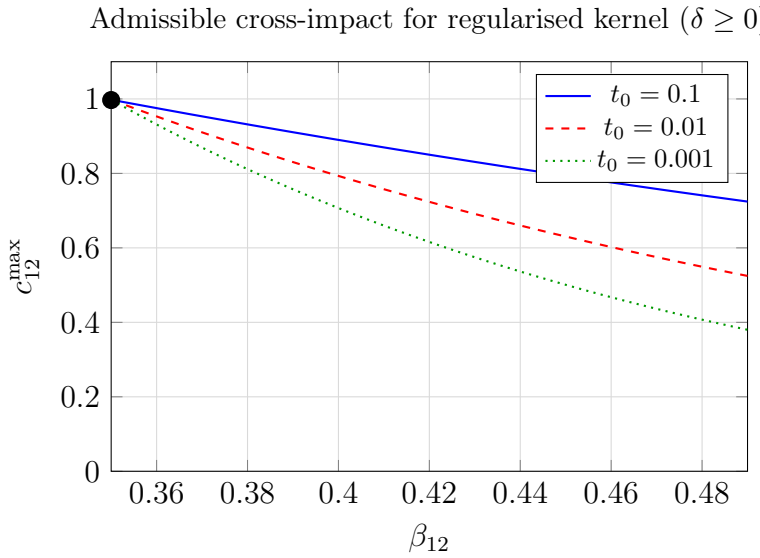


Figure 2: Admissible $c_{12}^{\max}$ for the regularised kernel as a function of $\beta_{12} \ge 0.35$ ($\delta \ge 0$), for three values of the regularisation parameter t_0. The black dot marks the singular kernel's admissible point at $\beta_{12} = 0.35$. As $t_0 \to 0$, the admissible region collapses to this point. Note: the plotted curves use approximate Gamma-ratio values; exact curves involve incomplete Gamma function ratios.

Figure 2 illustrates the key finding: for $\delta > 0$ (cross-impact decaying slower than the self-impact mean), the regularisation creates a nonzero admissible region whose width grows

with t_0 and with δ. For $\delta < 0$ (not shown), the admissible region remains $\{c_{12} = 0\}$ regardless of t_0—the regularisation provides no relief.

8.5 Admissible region for regularised kernel: t_0 dependence

Table 3 shows the maximum admissible $|c_{12}|$ for the regularised kernel at several (δ, t_0) combinations, computed from (20).

Table 3: Maximum admissible $|c_{12}|$ for the regularised kernel ($\beta_{11} = 0.3$, $\beta_{22} = 0.4$) as a function of δ and t_0.

δ	$t_0 = 0.1$	$t_0 = 0.01$	$t_0 = 0.001$	$t_0 \to 0$
-0.10	0 (necessity)	0	0	0
-0.05	0	0	0	0
0	0.997	0.997	0.997	0.997
0.02	0.954	0.923	0.893	0
0.05	0.891	0.809	0.734	0
0.10	0.793	0.630	0.501	0
0.20	0.629	0.397	0.250	0

The table confirms the asymmetric necessity structure:

- For $\delta < 0$: $c_{12}^{\max} = 0$ regardless of t_0 (mean-exponent condition is necessary).
- For $\delta = 0$: $c_{12}^{\max} = 0.997$ regardless of t_0 (this is the singular kernel's bound).
- For $\delta > 0$: $c_{12}^{\max}$ decreases as t_0 decreases, converging to 0. For fixed t_0, $c_{12}^{\max}$ decreases as δ increases (the exponent mismatch widens).

9 Conclusion

We have provided a complete characterisation of the no-price-manipulation condition for multi-asset markets with heterogeneous power-law cross-impact. The result reveals two fundamental constraints: the *mean-exponent condition*, which pins cross-impact decay to the arithmetic mean of self-impact decays, and *Gamma-weighted positive definiteness* of the coefficient matrix. These conditions are both necessary and sufficient for the singular kernel $H_{ij}(t) = c_{ij}\, t^{-\beta_{ij}}$.

For the regularised kernel $(t + t_0)^{-\beta_{ij}}$, we discovered an asymmetric necessity structure: the mean-exponent condition remains necessary when cross-impact decays faster than the self-impact mean, but not when it decays slower. This asymmetry, proved via spectral asymptotics of incomplete Gamma functions, reveals that the regularisation's effect on manipulation constraints is fundamentally one-sided. As $t_0 \to 0^+$, the admissible region for the relaxed case shrinks to zero, recovering the singular kernel's full necessity.

The boundary analysis revealed three further structural insights: (i) the permanent-impact limit $\beta \to 0^+$ recovers the Kyle condition $C \geq 0$, providing a clean interpolation between permanent and transient impact; (ii) heterogeneous exponents provide a structural safety margin at the PSD boundary by preventing zero-cost round trips; and (iii) positive

cross-impact scales to arbitrarily large portfolios, while negative cross-impact must vanish as $O(1/(N-1))$.

Several directions for future research remain.

1. **Full necessity for the regularised kernel.** Proposition 5.5 provides partial necessity for $N = 2$. Conjecture 5.8 proposes a natural extension to general N using the spectral matrix at $\omega = 0$, but the proof requires new techniques for the global minimum of the spectral ratio.
2. **Empirical testing of the mean-exponent condition.** The estimation methodology of Hey, Neuman, and Tuschmann [12] can be applied to test whether $\beta_{ij} \approx (b_i + b_j)/2$ holds in practice. Violations would indicate either model misspecification or latent manipulation opportunities.
3. **Stochastic liquidity.** Muhle-Karbe, Wang, and Webster [18] show that nonlinear impact converges to a linear propagator with stochastic liquidity. What are the no-manipulation conditions when c_{ij} and/or β_{ij} are stochastic processes?
4. **Nonlinear cross-impact.** The extension to concave cross-impact propagators (cf. [11]) and time-varying market conditions [19] presents natural challenges.
5. **The diffusivity boundary $\beta = 1/2$.** The borderline exponent $\beta = 1/2$ appears to saturate the "diffusivity condition" of Bouchaud et al. [6, 17]. Understanding the interplay between the mean-exponent condition and diffusivity at $\beta = 1/2$ remains open.
6. **Random matrix theory for Schur products.** A rigorous version of the large-N heuristic (27) would connect our results to the random matrix literature and provide practical bounds for portfolio managers.

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